Lab no. 3

Working with Git History, Stashing, and Reverting Commits

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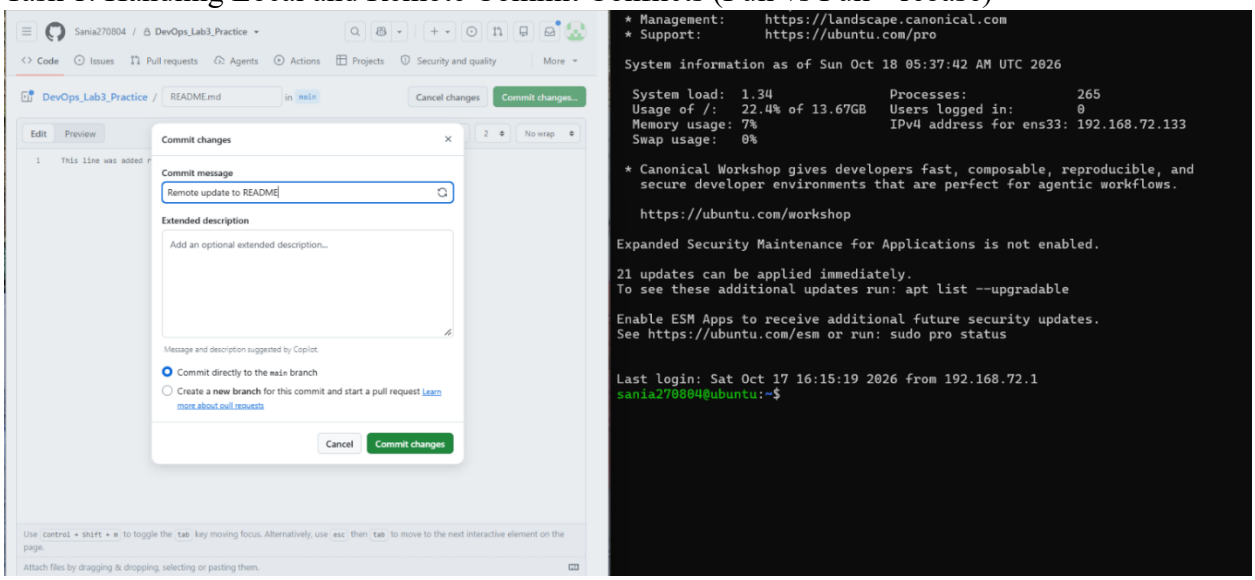
REGISTRATION NO.: 2023-BCS-078

COURSE CODE: CS-649

SECTION: A

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Task 1: Handling Local and Remote Commit Conflicts (Pull vs Pull --rebase)



Open your repository on **GitHub** and edit the README.md file directly in the browser.

```
sania270804@ubuntu:~/DevOps_Lab3_Practice$ nano README.md
sania270804@ubuntu:~/DevOps_Lab3_Practice$ git status
On branch main
Your branch is up to date with 'origin/main'.

Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   README.md

no changes added to commit (use "git add" and/or "git commit -a")
sania270804@ubuntu:~/DevOps_Lab3_Practice$ git add README.md
sania270804@ubuntu:~/DevOps_Lab3_Practice$ git commit -m "Local update to README"
[main c4ab848] Local update to README
 1 file changed, 1 insertion(+)
sania270804@ubuntu:~/DevOps_Lab3_Practice$
```

Edit the same README.md file locally.

```
sania270804@ubuntu:~/DevOps_Lab3_Practice$ git push origin main
To github.com:sania270804/DevOps_Lab3_Practice.git
 ! [rejected]        main -> main (fetch first)
error: failed to push some refs to 'github.com:sania270804/DevOps_Lab3_Practice.git'
hint: Updates were rejected because the remote contains work that you do not
hint: have locally. This is usually caused by another repository pushing to
hint: the same ref. If you want to integrate the remote changes, use
hint: 'git pull' before pushing again.
hint: See the 'Note about fast-forwards' in 'git push --help' for details.
sania270804@ubuntu:~/DevOps_Lab3_Practice$
```

Try to push.

```
sania270804@ubuntu:~/DevOps_Lab3_Practice$ git pull --no-rebase origin main
remote: Enumerating objects: 4, done.
remote: Counting objects: 100% (4/4), done.
remote: Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Unpacking objects: 100% (3/3), 934 bytes | 467.00 KiB/s, done.
From github.com:sania270804/DevOps_Lab3_Practice
 * branch            main       -> FETCH_HEAD
   d2b62c0..0969fed  main       -> origin/main
Auto-merging README.md
CONFLICT (add/add): Merge conflict in README.md
Automatic merge failed; fix conflicts and then commit the result.
sania270804@ubuntu:~/DevOps_Lab3_Practice$ nano README.md
sania270804@ubuntu:~/DevOps_Lab3_Practice$ git add README.md
sania270804@ubuntu:~/DevOps_Lab3_Practice$ git commit -m "Resolved merge conflict"
[main 93f2fbd] Resolved merge conflict
sania270804@ubuntu:~/DevOps_Lab3_Practice$
```

Git will merge your local and remote commits, creating a merge commit.

```
sania270804@ubuntu:~/DevOps_Lab3_Practice$ git push -u origin main
Enumerating objects: 9, done.
Counting objects: 100% (9/9), done.
Delta compression using up to 2 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (6/6), 594 bytes | 594.00 KiB/s, done.
Total 6 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote: This repository moved. Please use the new location:
remote:   git@github.com:Sania270804/DevOps_Lab3_Practice.git
To github.com:sania270804/DevOps_Lab3_Practice.git
   0969fed..93f2fbd  main -> main
branch 'main' set up to track 'origin/main'.
sania270804@ubuntu:~/DevOps_Lab3_Practice$
```

Push again.

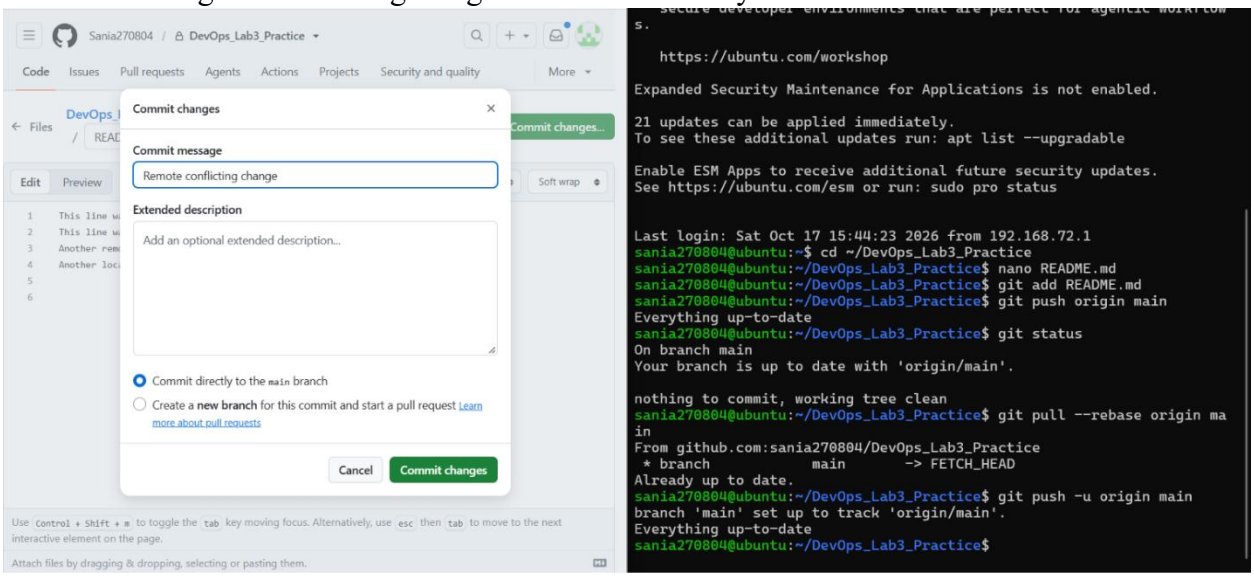
```
sania270804@ubuntu:~/DevOps_Lab3_Practice$ git pull --rebase origin main
From github.com:sania270804/DevOps_Lab3_Practice
 * branch            main       -> FETCH_HEAD
Auto-merging README.md
CONFLICT (content): Merge conflict in README.md
error: could not apply 1610637... Another Local update to README
hint: Resolve all conflicts manually, mark them as resolved with
hint: "git add/rm <conflicted_files>", then run "git rebase --continue".
hint: You can instead skip this commit: run "git rebase --skip".
hint: To abort and get back to the state before "git rebase", run "git rebase --abort".
hint: Disable this message with "git config set advice.mergeConflict false"
Could not apply 1610637... # Another Local update to README
sania270804@ubuntu:~/DevOps_Lab3_Practice$ nano README.md
sania270804@ubuntu:~/DevOps_Lab3_Practice$ git add README.md
sania270804@ubuntu:~/DevOps_Lab3_Practice$ git commit -m "Second Resolved merge conflict"
[detached HEAD fd7728b] Second Resolved merge conflict
 1 file changed, 2 insertions(+)
sania270804@ubuntu:~/DevOps_Lab3_Practice$
```

Git will replay your local commits on top of the pulled commits, keeping history linear.

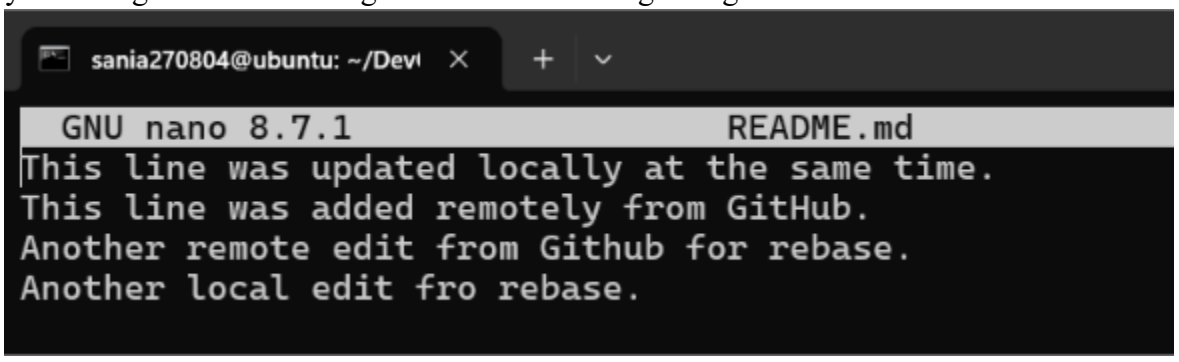
```
sania270804@ubuntu:~/DevOps_Lab3_Practice$ git push -u origin main
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 2 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 316 bytes | 316.00 KiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
remote: This repository moved. Please use the new location:
remote:   git@github.com:Sania270804/DevOps_Lab3_Practice.git
To github.com:sania270804/DevOps_Lab3_Practice.git
   2a224a0..0fff3f0  main -> main
branch 'main' set up to track 'origin/main'.
sania270804@ubuntu:~/DevOps_Lab3_Practice$
```

Push again.

Task 2: Creating and Resolving Merge Conflicts Manually



On GitHub (remote), open your README.md file and change an existing line.Commit your change with the message: Remote conflicting change



On your local machine, edit the same line in the same file but make a different change.

```
sania270804@ubuntu:~/DevOps_Lab3_Practice$ git add README.md
sania270804@ubuntu:~/DevOps_Lab3_Practice$ git commit -m "Local conflicting change"
[main f0a0836] Local conflicting change
 1 file changed, 1 insertion(+), 1 deletion(-)
sania270804@ubuntu:~/DevOps_Lab3_Practice$
```

Stage and commit your local change.

```
sania270804@ubuntu:~/DevOps_Lab3_Practice$ git push -u origin main
To github.com:sania270804/DevOps_Lab3_Practice.git
 ! [rejected]        main -> main (fetch first)
error: failed to push some refs to 'github.com:sania270804/DevOps_Lab3_Practice.git'
hint: Updates were rejected because the remote contains work that you do not
hint: have locally. This is usually caused by another repository pushing to
hint: the same ref. If you want to integrate the remote changes, use
hint: 'git pull' before pushing again.
hint: See the 'Note about fast-forwards' in 'git push --help' for details.
sania270804@ubuntu:~/DevOps_Lab3_Practice$
```

The push is rejected, since the remote version has conflicting changes.

```
sania270804@ubuntu:~/DevOps_Lab3_Practice$ git pull --rebase origin main
remote: Enumerating objects: 5, done.
remote: Counting objects: 100% (5/5), done.
remote: Compressing objects: 100% (2/2), done.
remote: Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Unpacking objects: 100% (3/3), 987 bytes | 987.00 KiB/s, done.
From github.com:sania270804/DevOps_Lab3_Practice
 * branch            main       -> FETCH_HEAD
   0fff3f0..6be78ae  main       -> origin/main
Auto-merging README.md
CONFLICT (content): Merge conflict in README.md
error: could not apply f0a0836... Local conflicting change
hint: Resolve all conflicts manually, mark them as resolved with
hint: "git add/rm <conflicted_files>", then run "git rebase --continue".
hint: You can instead skip this commit: run "git rebase --skip".
hint: To abort and get back to the state before "git rebase", run "git rebase --abort".
hint: Disable this message with "git config set advice.mergeConflict false"
Could not apply f0a0836... # Local conflicting change
sania270804@ubuntu:~/DevOps_Lab3_Practice$
```

Pull with rebase to bring in remote changes.

```
sania270804@ubuntu: ~/Devl
  GNU nano 8.7.1                                          README.md *
This line was added locally.
This line was updated remotely again.
This line was updated locally at the same time.
This line was added remotely from GitHub.
Another remote edit from Github for rebase.
Another local edit fro rebase.
```

Open the README.md file in your editor, you'll see conflict markers. Manually edit the file to keep the correct line (for example, merge both ideas).

```
sania270804@ubuntu:~/DevOps_Lab3_Practice$ git add README.md
sania270804@ubuntu:~/DevOps_Lab3_Practice$ git rebase --continue
[detached HEAD 50e217f] Local conflicting change
 1 file changed, 2 insertions(+)
Successfully rebased and updated refs/heads/main.
sania270804@ubuntu:~/DevOps_Lab3_Practice$
```

After fixing the file, mark the conflict as resolved and continue the rebase.

```
sania270804@ubuntu:~/DevOps_Lab3_Practice$ git push -u origin main
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 2 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 316 bytes | 316.00 KiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
remote: This repository moved. Please use the new location:
remote:   git@github.com:Sania270804/DevOps_Lab3_Practice.git
To github.com:sania270804/DevOps_Lab3_Practice.git
   2a224a0..0fff3f0  main -> main
branch 'main' set up to track 'origin/main'.
sania270804@ubuntu:~/DevOps_Lab3_Practice$
```

Finally, push your changes.

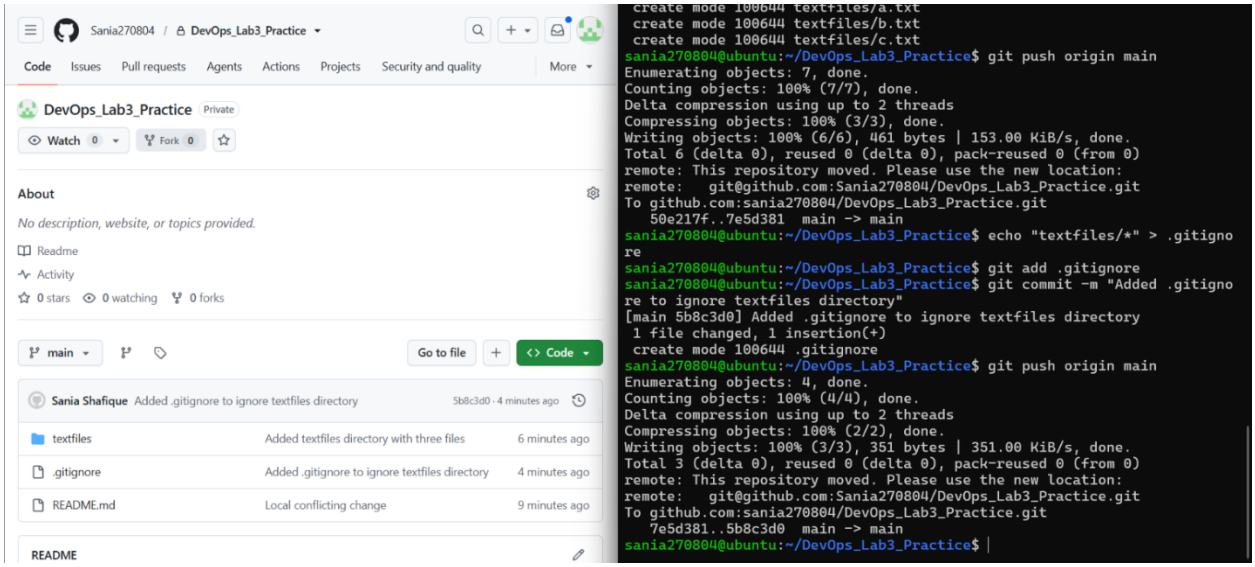
Task 3: Managing Ignored Files with .gitignore and Removing Tracked Files

```
sania270804@ubuntu:~/DevOps_Lab3_Practice$ mkdir textfiles
sania270804@ubuntu:~/DevOps_Lab3_Practice$ echo "File A content" > textfiles/a.txt
sania270804@ubuntu:~/DevOps_Lab3_Practice$ echo "File B content" > textfiles/b.txt
sania270804@ubuntu:~/DevOps_Lab3_Practice$ echo "File C content" > textfiles/c.txt
sania270804@ubuntu:~/DevOps_Lab3_Practice$ git add .
sania270804@ubuntu:~/DevOps_Lab3_Practice$ git commit -m "Added textfiles directory with three files"
[main 7e5d381] Added textfiles directory with three files
 3 files changed, 3 insertions(+)
 create mode 100644 textfiles/a.txt
 create mode 100644 textfiles/b.txt
 create mode 100644 textfiles/c.txt
sania270804@ubuntu:~/DevOps_Lab3_Practice$ git push origin main
Enumerating objects: 7, done.
Counting objects: 100% (7/7), done.
Delta compression using up to 2 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (6/6), 461 bytes | 153.00 KiB/s, done.
Total 6 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote: This repository moved. Please use the new location:
remote:   git@github.com:Sania270804/DevOps_Lab3_Practice.git
To github.com:sania270804/DevOps_Lab3_Practice.git
   50e217f..7e5d381  main -> main
sania270804@ubuntu:~/DevOps_Lab3_Practice$
```

Create a new folder named textfiles inside your repository. Inside the textfiles folder, create three text files. Add and commit the new directory.

```
sania270804@ubuntu:~/DevOps_Lab3_Practice$ echo "textfiles/*" > .gitignore
sania270804@ubuntu:~/DevOps_Lab3_Practice$ git add .gitignore
sania270804@ubuntu:~/DevOps_Lab3_Practice$ git commit -m "Added .gitignore to ignore textfiles directory"
[main 5b8c3d0] Added .gitignore to ignore textfiles directory
 1 file changed, 1 insertion(+)
 create mode 100644 .gitignore
sania270804@ubuntu:~/DevOps_Lab3_Practice$ git push origin main
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 2 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 351 bytes | 351.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote: This repository moved. Please use the new location:
remote:   git@github.com:Sania270804/DevOps_Lab3_Practice.git
To github.com:sania270804/DevOps_Lab3_Practice.git
   7e5d381..5b8c3d0  main -> main
sania270804@ubuntu:~/DevOps_Lab3_Practice$
```

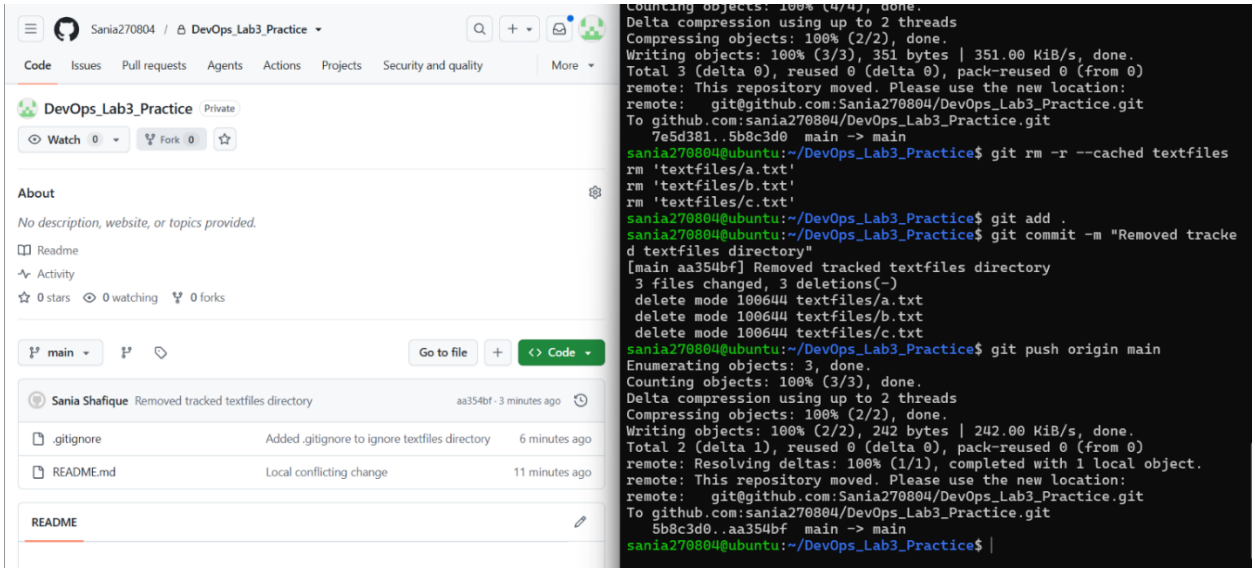
Now, create a .gitignore file in the root of your repository. Add and commit the .gitignore file.



Go to GitHub and check your repository, notice that the textfiles directory is still visible on the remote.

```
sania270804@ubuntu:~/DevOps_Lab3_Practice$ git rm -r --cached textfiles
rm 'textfiles/a.txt'
rm 'textfiles/b.txt'
rm 'textfiles/c.txt'
sania270804@ubuntu:~/DevOps_Lab3_Practice$ git add .
sania270804@ubuntu:~/DevOps_Lab3_Practice$ git commit -m "Removed tracked textfiles directory"
[main aa354bf] Removed tracked textfiles directory
 3 files changed, 3 deletions(-)
 delete mode 100644 textfiles/a.txt
 delete mode 100644 textfiles/b.txt
 delete mode 100644 textfiles/c.txt
sania270804@ubuntu:~/DevOps_Lab3_Practice$ git push origin main
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Delta compression using up to 2 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (2/2), 242 bytes | 242.00 KiB/s, done.
Total 2 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
remote: This repository moved. Please use the new location:
remote:   git@github.com:Sania270804/DevOps_Lab3_Practice.git
To github.com:sania270804/DevOps_Lab3_Practice.git
   5b8c3d0..aa354bf  main -> main
sania270804@ubuntu:~/DevOps_Lab3_Practice$
```

Remove already tracked files from Git. Commit and push again.



Check your GitHub repository again, the textfiles folder should now be deleted remotely.

Task 4: Create Temporary Changes and Use git stash

```
sania270804@ubuntu:~/DevOps_Lab3_Practice$ git checkout -b feature-branch
Switched to a new branch 'feature-branch'
sania270804@ubuntu:~/DevOps_Lab3_Practice$ nano README.md
sania270804@ubuntu:~/DevOps_Lab3_Practice$ git add README.md
sania270804@ubuntu:~/DevOps_Lab3_Practice$ git commit -m "Changes the README.md file"
[feature-branch cbfa5b1] Changes the README.md file
 1 file changed, 1 insertion(+), 1 deletion(-)
sania270804@ubuntu:~/DevOps_Lab3_Practice$
```

Modify any file (for example, README.md) by adding a few test lines.

```
sania270804@ubuntu:~/DevOps_Lab3_Practice$ nano README.md
sania270804@ubuntu:~/DevOps_Lab3_Practice$ git checkout main
error: Your local changes to the following files would be overwritten by checkout:
        README.md
Please commit your changes or stash them before you switch branches.
Aborting
sania270804@ubuntu:~/DevOps_Lab3_Practice$
```

Without committing or stashing the changes, try to switch to another branch.

```
sania270804@ubuntu:~/DevOps_Lab3_Practice$ git stash
Saved working directory and index state WIP on feature-branch: cbfa5b1 Changes the README.md file
sania270804@ubuntu:~/DevOps_Lab3_Practice$
```

To fix this, stash your changes.

```
sania270804@ubuntu:~/DevOps_Lab3_Practice$ git checkout main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.
sania270804@ubuntu:~/DevOps_Lab3_Practice$
```

Try switching branches again, it should now work.

```
sania270804@ubuntu:~/DevOps_Lab3_Practice$ git checkout feature-branch
Switched to branch 'feature-branch'
sania270804@ubuntu:~/DevOps_Lab3_Practice$
```

Return to your previous branch.

```
sania270804@ubuntu:~/DevOps_Lab3_Practice$ git status
On branch feature-branch
nothing to commit, working tree clean
sania270804@ubuntu:~/DevOps_Lab3_Practice$
```

Check your working directory status.

```
sania270804@ubuntu:~/DevOps_Lab3_Practice$ git stash pop
On branch feature-branch
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   README.md

no changes added to commit (use "git add" and/or "git commit -a")
Dropped refs/stash@{0} (9fa53c636191a6fb5986f6e9648914890ea138f8)
sania270804@ubuntu:~/DevOps_Lab3_Practice$
```

Now restore your stashed changes.

Task 5: Checkout a Specific Commit Using git log

```
sania270804@ubuntu:~/DevOps_Lab3_Practice$ git log --oneline
cbfa5b1 (HEAD -> feature-branch) Changes the README.md file
aa354bf (origin/main, origin/HEAD, main) Removed tracked textfiles directory
5b8c3d0 Added .gitignore to ignore textfiles directory
7e5d381 Added textfiles directory with three files
50e217f Local conflicting change
6be78ae Remote conflicting change
0fff3f0 Second Resolved merge conflict
2a224a0 Add remote edit note for rebase
93f2fbd Resolved merge conflict
a86a686 Local update to README
0969fed Remote update to README
d2b62c0 Delete README.md
ccd7995 Local update to README
b36f319 Resolved merge conflict
a003cf8 Local update to README
c311a2f Remote update to README
14ac2ad Delete README.md
54261c5 Remote update to README
85bcbda Delete README.md
c4ab848 Local update to README
014b4e9 Remote update to README
052091c Create README.md
```

View commit history.

```
sania270804@ubuntu:~/DevOps_Lab3_Practice$ git checkout cbfa5b1
M       README.md
Note: switching to 'cbfa5b1'.

You are in 'detached HEAD' state. You can look around, make experimental
changes and commit them, and you can discard any commits you make in this
state without impacting any branches by switching back to a branch.

If you want to create a new branch to retain commits you create, you may
do so (now or later) by using -c with the switch command. Example:

  git switch -c <new-branch-name>

Or undo this operation with:

  git switch -

Turn off this advice by setting config variable advice.detachedHead to false

HEAD is now at cbfa5b1 Changes the README.md file
sania270804@ubuntu:~/DevOps_Lab3_Practice$
```

Checkout the commit.

```
sania270804@ubuntu:~/DevOps_Lab3_Practice$ git checkout main
Previous HEAD position was cbfa5b1 Changes the README.md file
Switched to branch 'main'
Your branch is up to date with 'origin/main'.
sania270804@ubuntu:~/DevOps_Lab3_Practice$
```

Return to your main branch.

Task 6: Resetting Commits (Soft vs Hard Reset)

```
sania270804@ubuntu:~/DevOps_Lab3_Practice$ nano README.md
sania270804@ubuntu:~/DevOps_Lab3_Practice$ git add .
sania270804@ubuntu:~/DevOps_Lab3_Practice$ git commit -m "Added test line"
[main 32345f4] Added test line
 1 file changed, 1 insertion(+), 1 deletion(-)
sania270804@ubuntu:~/DevOps_Lab3_Practice$
```

Add a new line in any file and commit it.

```
sania270804@ubuntu:~/DevOps_Lab3_Practice$ nano README.md
sania270804@ubuntu:~/DevOps_Lab3_Practice$ git add .
sania270804@ubuntu:~/DevOps_Lab3_Practice$ git commit -m "Second test commit"
[main 67c2ee7] Second test commit
 1 file changed, 1 insertion(+)
sania270804@ubuntu:~/DevOps_Lab3_Practice$
```

Add another change and commit again.

```
sania270804@ubuntu:~/DevOps_Lab3_Practice$ git log --oneline
67c2ee7 (HEAD -> main) Second test commit
32345f4 Added test line
aa354bf (origin/main, origin/HEAD) Removed tracked textfiles directory
5b8c3d0 Added .gitignore to ignore textfiles directory
7e5d381 Added textfiles directory with three files
50e217f Local conflicting change
6be78ae Remote conflicting change
0fff3f0 Second Resolved merge conflict
2a224a0 Add remote edit note for rebase
93f2fbd Resolved merge conflict
a86a686 Local update to README
0969fed Remote update to README
d2b62c0 Delete README.md
ccd7995 Local update to README
b36f319 Resolved merge conflict
a003cf8 Local update to README
c311a2f Remote update to README
14ac2ad Delete README.md
54261c5 Remote update to README
85bcbda Delete README.md
c4ab848 Local update to README
014b4e9 Remote update to README
052091c Create README.md
sania270804@ubuntu:~/DevOps_Lab3_Practice$
```

View the history before reset.

```
sania270804@ubuntu: ~/Dev
  GNU nano 8.7.1                                          README.md
This line was added locally.
This line was updated remotely again.
This line was updated locally at the same time.
This line was added remotely from GitHub.
Another remote edit from Github for rebase.
Another local edit fro rebase.
Adding new line for task 6.
Adding another change for task 6.
```

Check file contents for both changes.

```
sania270804@ubuntu:~/DevOps_Lab3_Practice$ git reset --soft HEAD~1
sania270804@ubuntu:~/DevOps_Lab3_Practice$
```

Perform a soft reset (keeps changes in working directory).

```
sania270804@ubuntu:~/DevOps_Lab3_Practice$ git log --oneline
32345f4 (HEAD -> main) Added test line
aa354bf (origin/main, origin/HEAD) Removed tracked textfiles directory
5b8c3d0 Added .gitignore to ignore textfiles directory
7e5d381 Added textfiles directory with three files
50e217f Local conflicting change
6be78ae Remote conflicting change
0fff3f0 Second Resolved merge conflict
2a224a0 Add remote edit note for rebase
93f2fbd Resolved merge conflict
a86a686 Local update to README
0969fed Remote update to README
d2b62c0 Delete README.md
ccd7995 Local update to README
b36f319 Resolved merge conflict
a003cf8 Local update to README
c311a2f Remote update to README
14ac2ad Delete README.md
54261c5 Remote update to README
85bcbda Delete README.md
c4ab848 Local update to README
014b4e9 Remote update to README
052091c Create README.md
sania270804@ubuntu:~/DevOps_Lab3_Practice$
```

Check commit history after soft reset.

```
sania270804@ubuntu: ~/Devl
  GNU nano 8.7.1                                          README.md
This line was added locally.
This line was updated remotely again.
This line was updated locally at the same time.
This line was added remotely from GitHub.
Another remote edit from Github for rebase.
Another local edit fro rebase.
Adding new line for task 6.
Adding another change for task 6.
```

Verify changes in the file.

```
sania270804@ubuntu: ~/Devl
  GNU nano 8.7.1                                          README.md
This line was added locally.
This line was updated remotely again.
This line was updated locally at the same time.
This line was added remotely from GitHub.
Another remote edit from Github for rebase.
Another local edit fro rebase.
Adding new line for task 6.
```

You should see your changes are staged (ready to commit again). Perform commit.

```
sania270804@ubuntu:~/DevOps_Lab3_Practice$ git reset --hard HEAD~1
HEAD is now at 32345f4 Added test line
sania270804@ubuntu:~/DevOps_Lab3_Practice$
```

Perform a hard reset (discards changes completely).

```
sania270804@ubuntu:~/DevOps_Lab3_Practice$ git log --oneline
32345f4 (HEAD -> main) Added test line
aa354bf (origin/main, origin/HEAD) Removed tracked textfiles directory
5b8c3d0 Added .gitignore to ignore textfiles directory
7e5d381 Added textfiles directory with three files
50e217f Local conflicting change
6be78ae Remote conflicting change
0fff3f0 Second Resolved merge conflict
2a224a0 Add remote edit note for rebase
93f2fbd Resolved merge conflict
a86a686 Local update to README
0969fed Remote update to README
d2b62c0 Delete README.md
ccd7995 Local update to README
b36f319 Resolved merge conflict
a003cf8 Local update to README
c311a2f Remote update to README
14ac2ad Delete README.md
54261c5 Remote update to README
85bcbda Delete README.md
c4ab848 Local update to README
014b4e9 Remote update to README
052091c Create README.md
sania270804@ubuntu:~/DevOps_Lab3_Practice$
```

Check commit history after hard reset.

```
sania270804@ubuntu: ~/Devl
  GNU nano 8.7.1                                        README.md
This line was added locally.
This line was updated remotely again.
This line was updated locally at the same time.
This line was added remotely from GitHub.
Another remote edit from Github for rebase.
Another local edit fro rebase.
Adding new line for task 6.
```

Open the file and confirm the second edit is gone (reverted to previous commit).

Task 7: Amending the Last Commit

```
sania270804@ubuntu:~/DevOps_Lab3_Practice$ nano README.md
sania270804@ubuntu:~/DevOps_Lab3_Practice$ git add .
sania270804@ubuntu:~/DevOps_Lab3_Practice$ git commit -m "Fix log message"
[main d8dde46] Fix log message
 1 file changed, 1 insertion(+)
sania270804@ubuntu:~/DevOps_Lab3_Practice$
```

Make a small change in any file. Stage it and commit:

```
sania270804@ubuntu:~/DevOps_Lab3_Practice$ nano README.md
sania270804@ubuntu:~/DevOps_Lab3_Practice$ git add .
sania270804@ubuntu:~/DevOps_Lab3_Practice$ git commit -m "Second test commit"
[main 67c2ee7] Second test commit
 1 file changed, 1 insertion(+)
sania270804@ubuntu:~/DevOps_Lab3_Practice$
```

Realize you forgot to change another file. Update it now. Add it and amend the last commit.

Task 8: Reverting a Commit (Safe Undo on Remote Branch)

```
sania270804@ubuntu:~/DevOps_Lab3_Practice$ echo "temporary text" >> temp.txt
sania270804@ubuntu:~/DevOps_Lab3_Practice$ git add .
sania270804@ubuntu:~/DevOps_Lab3_Practice$ git commit -m "Added temporary text file"
[main bafb72c] Added temporary text file
 1 file changed, 1 insertion(+)
 create mode 100644 temp.txt
sania270804@ubuntu:~/DevOps_Lab3_Practice$ git push origin main
Enumerating objects: 12, done.
Counting objects: 100% (12/12), done.
Delta compression using up to 2 threads
Compressing objects: 100% (8/8), done.
Writing objects: 100% (10/10), 933 bytes | 311.00 KiB/s, done.
Total 10 (delta 3), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (3/3), completed with 1 local object.
remote: This repository moved. Please use the new location:
remote:   git@github.com:Sania270804/DevOps_Lab3_Practice.git
To github.com:sania270804/DevOps_Lab3_Practice.git
   aa354bf..bafb72c  main -> main
sania270804@ubuntu:~/DevOps_Lab3_Practice$
```

Make a change and commit it.

```
sania270804@ubuntu:~/DevOps_Lab3_Practice$ git log --oneline
bafb72c (HEAD, origin/main, origin/HEAD, main) Added temporary text file
407deba Fix log message Another small change
32345f4 Added test line
aa354bf Removed tracked textfiles directory
5b8c3d0 Added .gitignore to ignore textfiles directory
7e5d381 Added textfiles directory with three files
50e217f Local conflicting change
6be78ae Remote conflicting change
0fff3f0 Second Resolved merge conflict
2a224a0 Add remote edit note for rebase
93f2fbd Resolved merge conflict
a86a686 Local update to README
0969fed Remote update to README
d2b62c0 Delete README.md
ccd7995 Local update to README
b36f319 Resolved merge conflict
a003cf8 Local update to README
c311a2f Remote update to README
14ac2ad Delete README.md
54261c5 Remote update to README
85bcbda Delete README.md
c4ab848 Local update to README
014b4e9 Remote update to README
052091c Create README.md
sania270804@ubuntu:~/DevOps_Lab3_Practice$ git revert bafb72c
[detached HEAD 5cde159] Revert "Added temporary text file"
 1 file changed, 1 deletion(-)
 delete mode 100644 temp.txt
sania270804@ubuntu:~/DevOps_Lab3_Practice$
```

Now revert that commit safely (do not delete it).

```
sania270804@ubuntu:~/DevOps_Lab3_Practice$ git push origin main
Everything up-to-date
sania270804@ubuntu:~/DevOps_Lab3_Practice$
```

Push the revert commit.

Task 9: Force Push (With Caution)

```
sania270804@ubuntu:~/DevOps_Lab3_Practice$ git checkout -b test-force
Switched to a new branch 'test-force'
sania270804@ubuntu:~/DevOps_Lab3_Practice$
```

Create a new branch.

```
sania270804@ubuntu:~/DevOps_Lab3_Practice$ echo "Testing force push" >> README.md
sania270804@ubuntu:~/DevOps_Lab3_Practice$ git add README.md
sania270804@ubuntu:~/DevOps_Lab3_Practice$ git commit -m "Testing force push commit"
[test-force 4a45c1b] Testing force push commit
 1 file changed, 1 insertion(+)
sania270804@ubuntu:~/DevOps_Lab3_Practice$
```

Make and commit a small change.

```
sania270804@ubuntu:~/DevOps_Lab3_Practice$ git push origin test-force
Enumerating objects: 7, done.
Counting objects: 100% (7/7), done.
Delta compression using up to 2 threads
Compressing objects: 100% (5/5), done.
Writing objects: 100% (5/5), 598 bytes | 598.00 KiB/s, done.
Total 5 (delta 2), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (2/2), completed with 2 local objects.
remote: This repository moved. Please use the new location:
remote:   git@github.com:Sania270804/DevOps_Lab3_Practice.git
remote:
remote: Create a pull request for 'test-force' on GitHub by visiting:
remote:      https://github.com/Sania270804/DevOps_Lab3_Practice/pull/new/test-force
remote:
To github.com:sania270804/DevOps_Lab3_Practice.git
 * [new branch]      test-force -> test-force
sania270804@ubuntu:~/DevOps_Lab3_Practice$
```

Push it.

```
sania270804@ubuntu:~/DevOps_Lab3_Practice$ git reset --hard HEAD~1
HEAD is now at 5cde159 Revert "Added temporary text file"
sania270804@ubuntu:~/DevOps_Lab3_Practice$
```

Perform a hard reset.

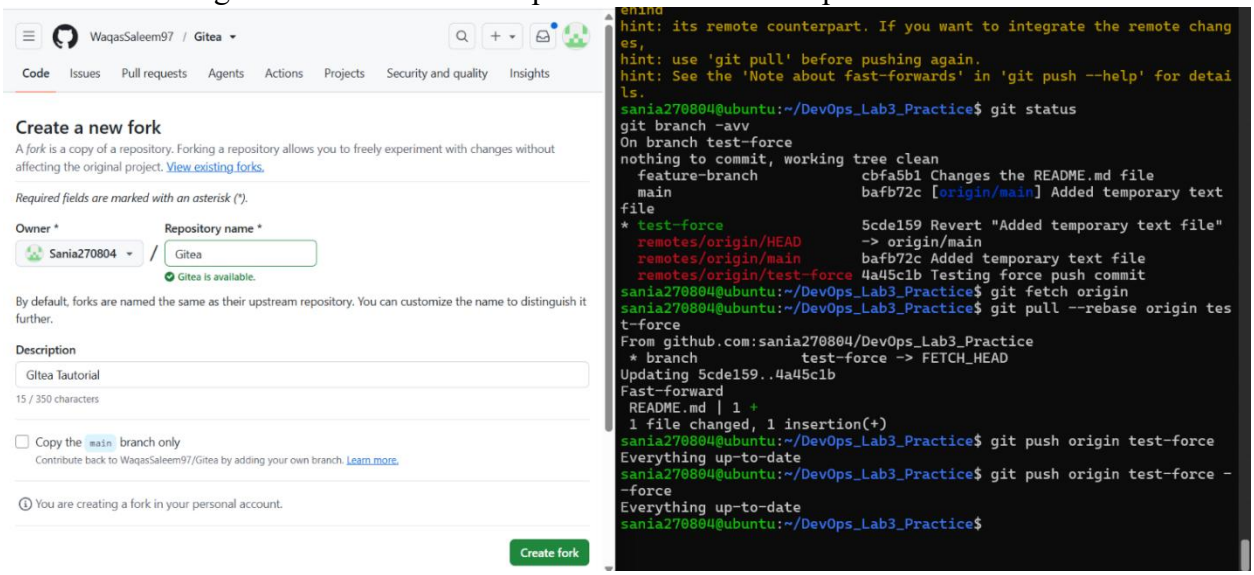
```
sania270804@ubuntu:~/DevOps_Lab3_Practice$ git push origin test-force
To github.com:sania270804/DevOps_Lab3_Practice.git
 ! [rejected]        test-force -> test-force (non-fast-forward)
error: failed to push some refs to 'github.com:sania270804/DevOps_Lab3_Practice.git'
hint: Updates were rejected because the tip of your current branch is behind
hint: its remote counterpart. If you want to integrate the remote changes,
hint: use 'git pull' before pushing again.
hint: See the 'Note about fast-forwards' in 'git push --help' for details.
```

Push again.

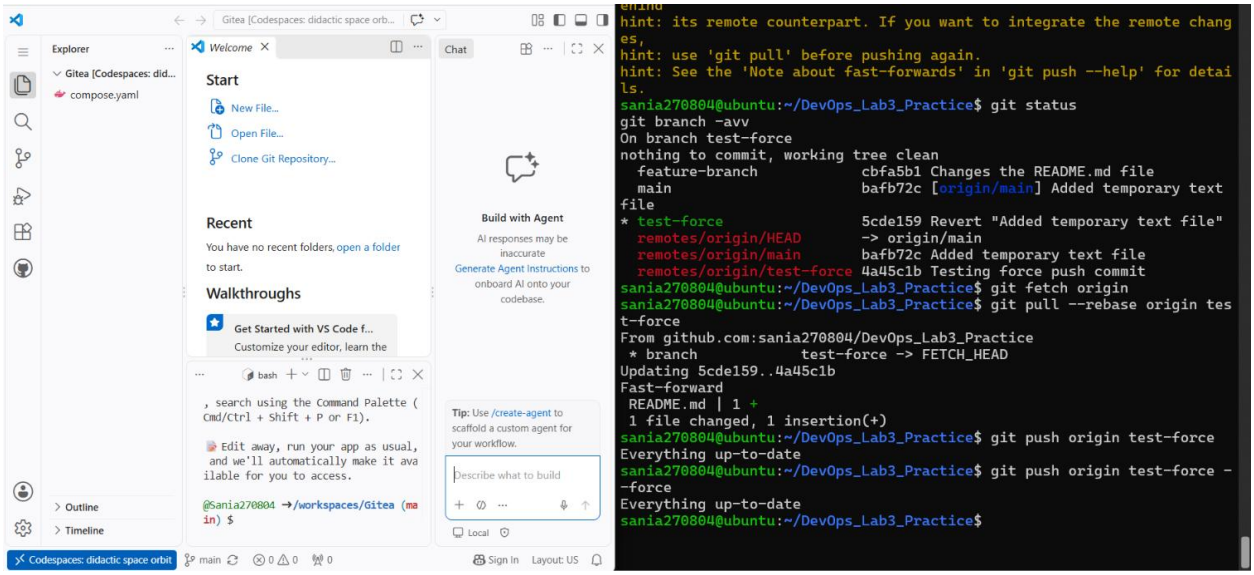
```
sania270804@ubuntu:~/DevOps_Lab3_Practice$ git push origin test-force --force
Everything up-to-date
```

Now force-push to remote.

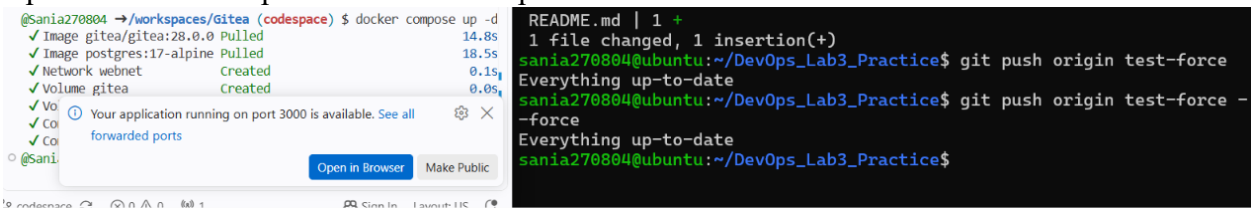
Task 10: Running Gitea in GitHub Codespaces via Docker Compose



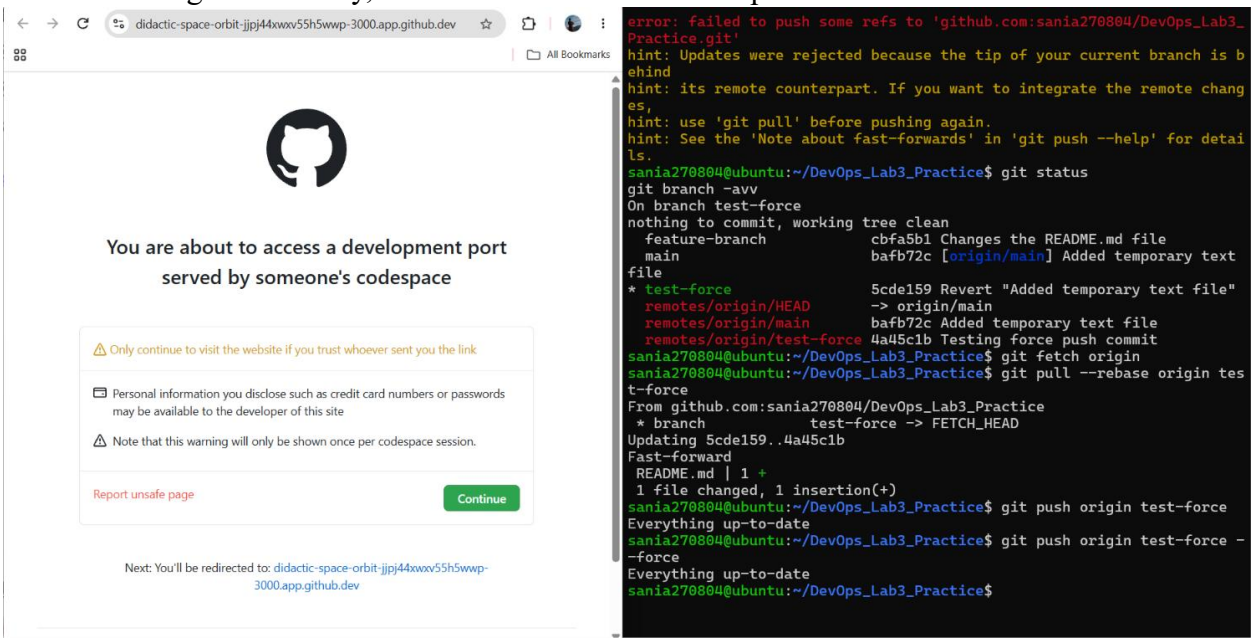
Fork the Gitea Repository.



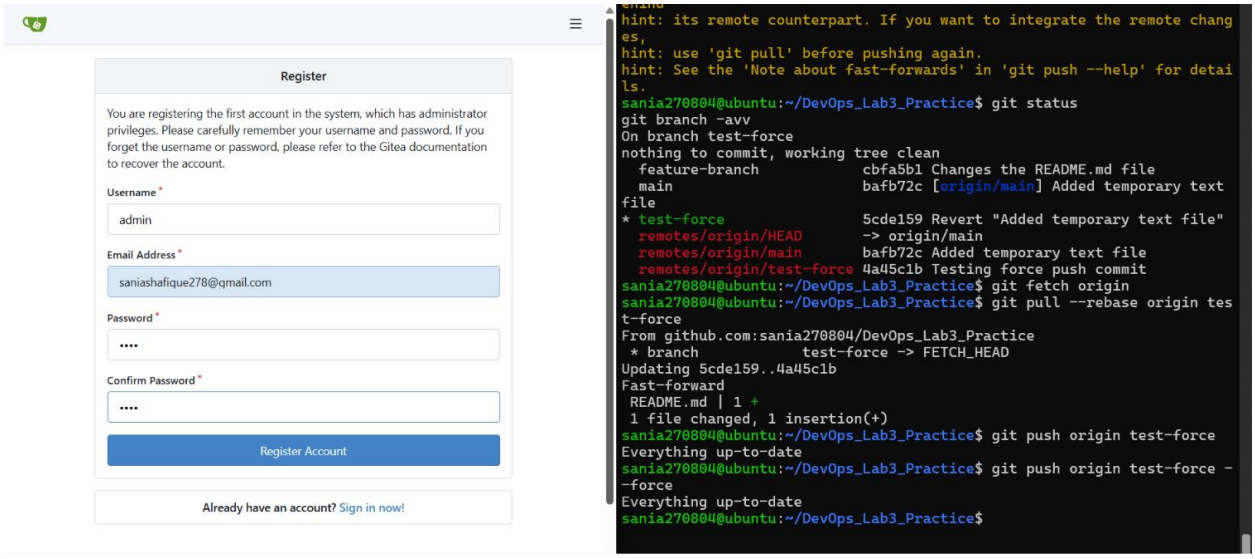
Open the Forked Repo in GitHub Codespaces.



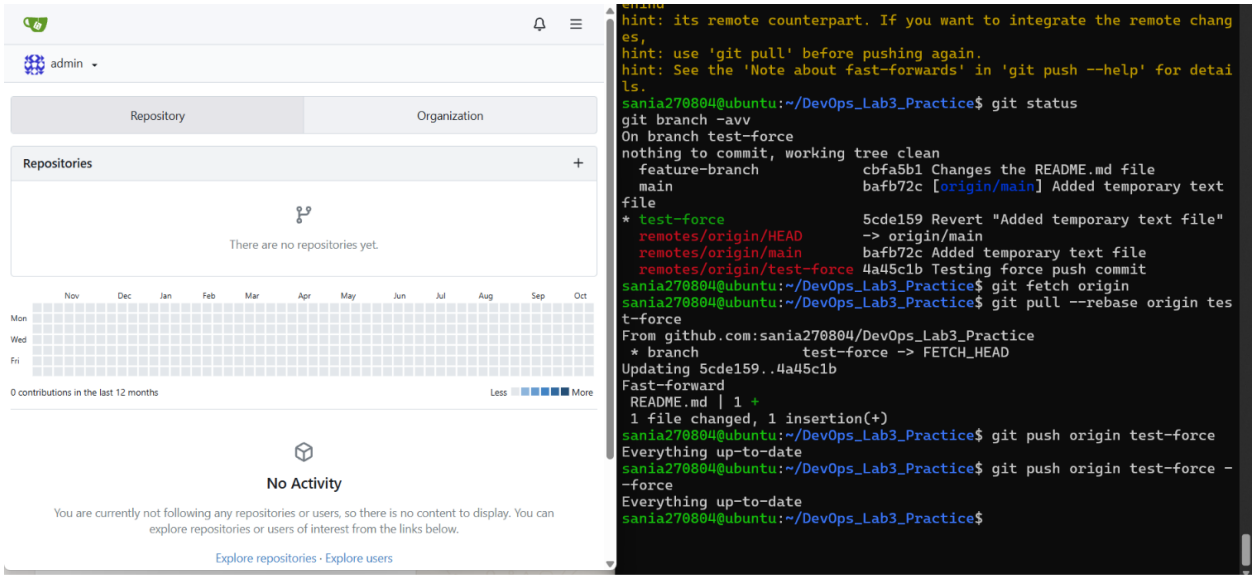
After switching successfully, start Gitea with Docker Compose.



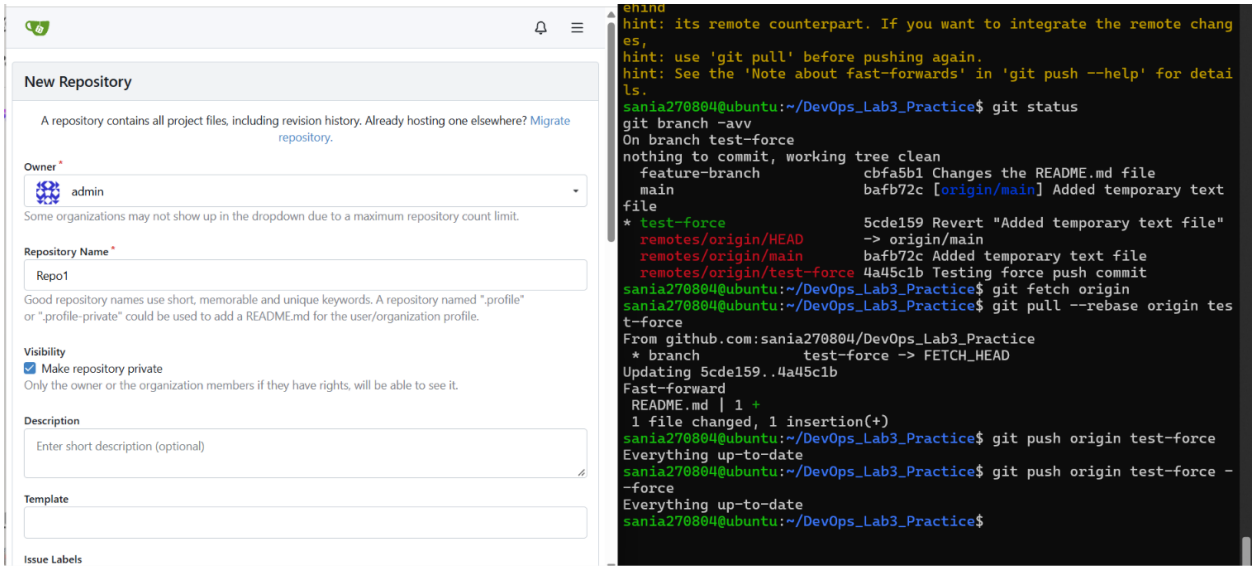
Access Gitea Web Interface.



Install Gitea.

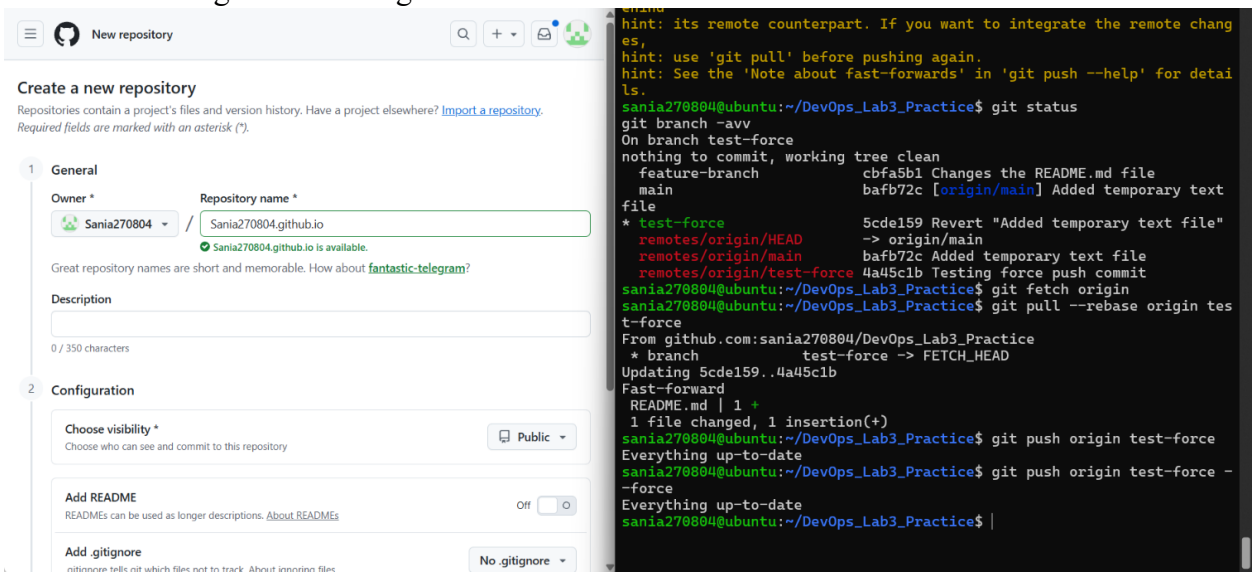


Log In to Gitea.

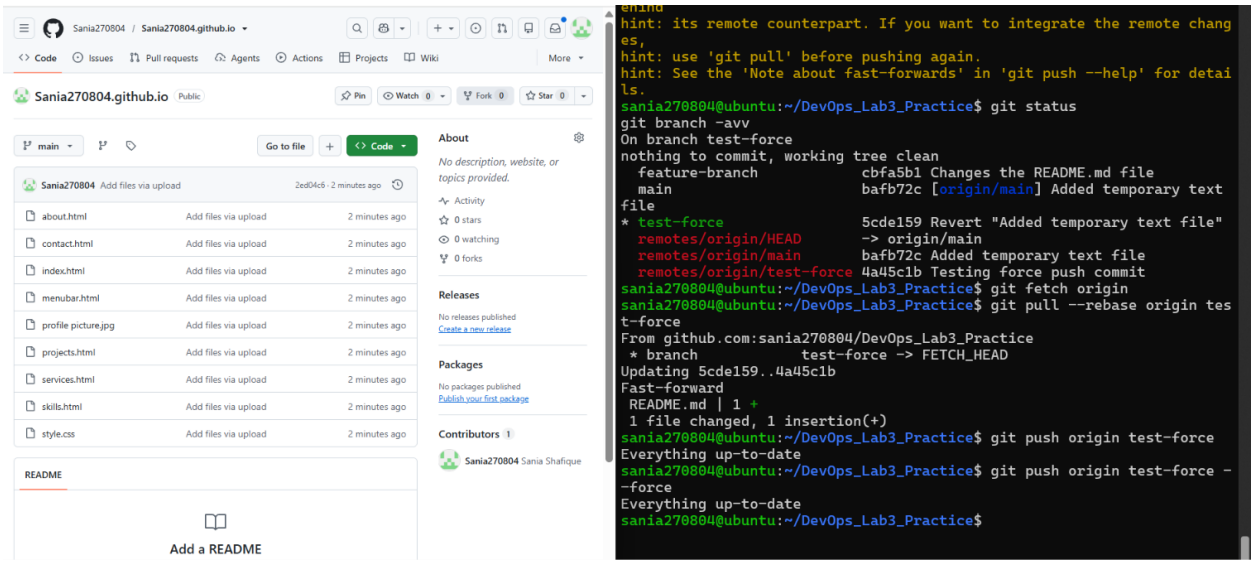


Create a New Repository in Gitea.

Task 11: Creating a GitHub Pages Portfolio Site



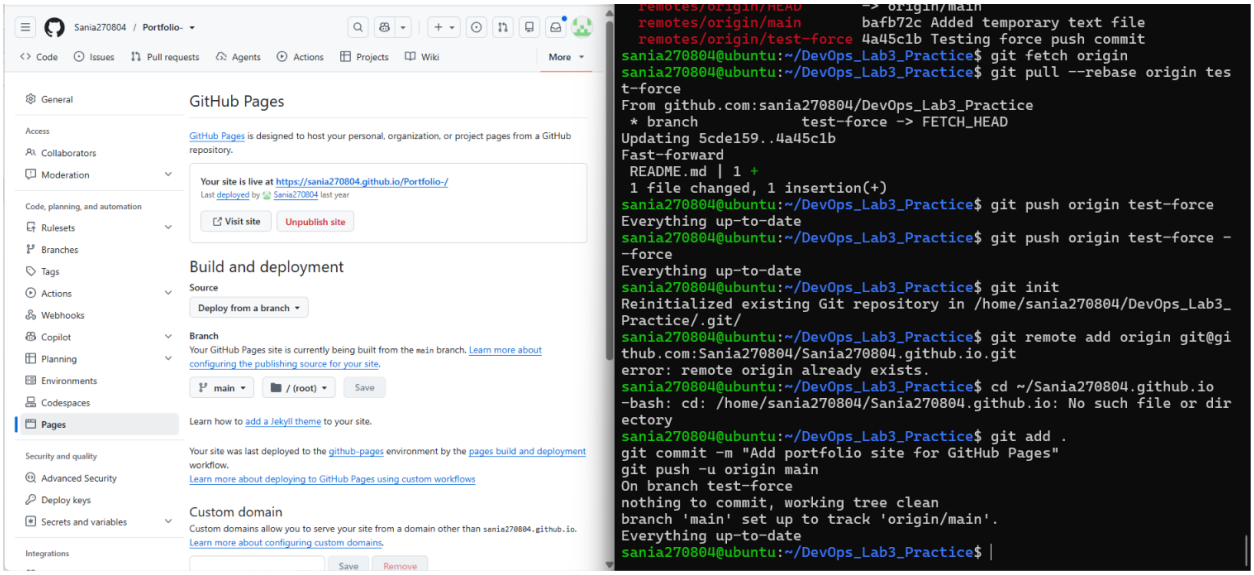
Create a GitHub Pages Repository.



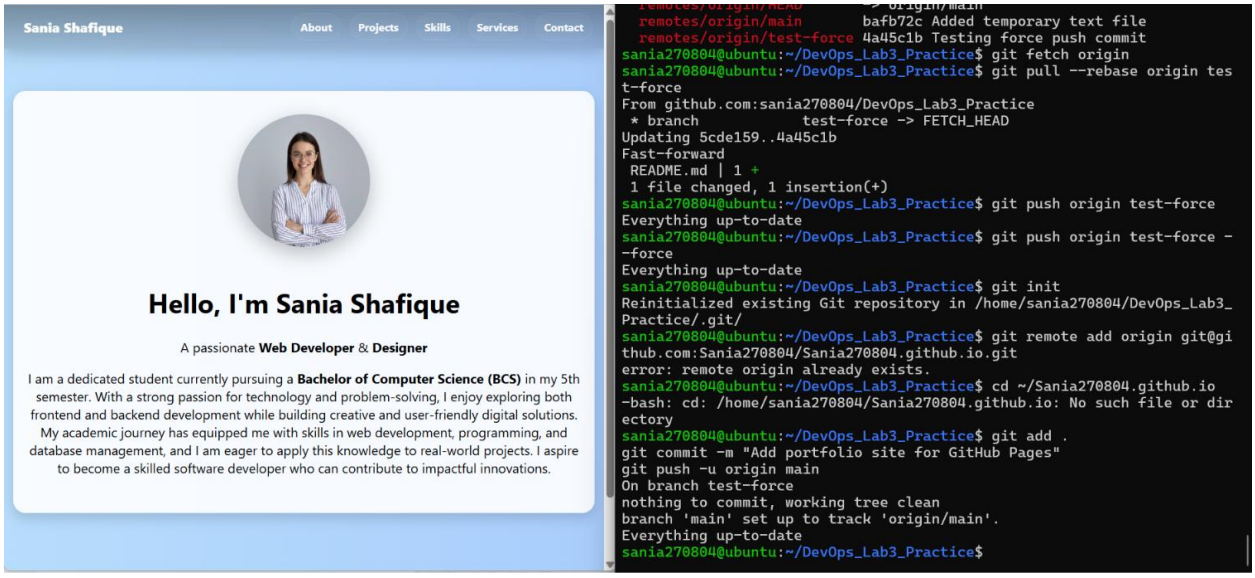
Add Static Website Code.



Push the Files to GitHub.



Check GitHub Pages Settings.



Visit Your Live Site.